Understanding the t-test

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Abstract

This report discusses Student's t-distribution. We begin by showing that test statistics computed using sample means and sample variances calculated from n normal random variables are not samples from a standard normal distribution. We then assume that this statistic is a sample from Student's t-distribution with $(n-1)$ degrees of freedom and show how the probability density function for this distribution approaches that of the normal distribution as n increases.

1 Introduction

When you do a hypothesis test to compare a sample mean, $\overline{X}$, with μ you will not typically have the true population variance. Consequently, we have to use the n pieces of data, X_i, in the sample to estimate the sample variance. The test statistic that is used in these hypothesis tests is thus computed using:

$$T = \frac{\overline{X} - \mu}{\sqrt{S^2/n}} \quad \text{where} \quad \overline{X} = \frac{1}{n}\sum_{i=1}^n X_i \quad \text{and} \quad S^2 = \frac{n}{n-1}\left[\frac{1}{n}\left(\sum_{i=1}^n X_i^2 \right) - \overline{X}^2 \right] \tag{1}$$

Most textbooks will tell you that when n is large this statistic can be assumed to be a sample from a standard normal random variable if the null hypothesis is assumed true. However, when n is small this approximation is invalid and one should instead assume that this is a sample from a Student's t-distribution with $(n-1)$ degrees of freedom. Very rarely do those textbooks explain the particular values of n that are large enough for you to use a normal distribution in these hypothesis tests. In this report we will thus investigate the t-distribution in an attempt to answer this question.

We begin the report by constructing a histogram by sampling the statistic defined in equation 1. We then compare this histogram with the probability density function for a standard normal distribution and a t-distribution and show that this statistic is **not** sampling from a standard normal distribution. We then show that the t-distribution does become progressively more and more similar to the standard normal distribution as the number of degrees of freedom is increased. However, our ultimate advice in the conclusion is to use the t-distribution when performing hypothesis tests using the test statistic in equation 1 with any

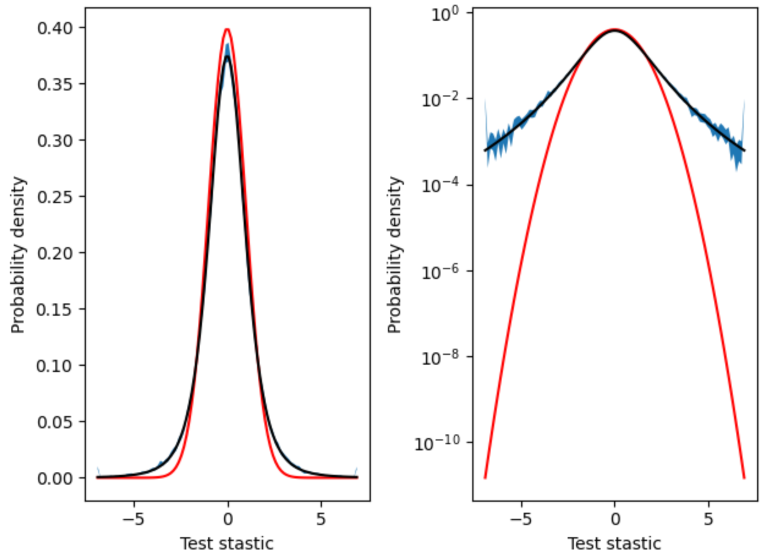

Fig. 1. An estimate for the probability density function for the statistic that is defined in equation 1 with $n = 5$. The blue shaded error shows the 90 % confidence limit on the estimate for the probability density that was obtained by sampling. The black and red lines are the probability density functions for a Student t-distribution with 4 degrees of freedom and a standard normal random variable. In the left panel the probability densities are plotted with on a linear y-scale, while on the right the y-axis is logarithmic. Plotting the probability densities in this second way makes the differences for the tails of the normal and t-distributions more apparent.

n value as this is the distribution sampled in the experiment. Modern computational statistics packages make it easy to compute the probability density function for a t-distribution with any number of degrees of freedom so there is not need for approximations.

2 Results and discussion

To demonstrate that the t-distribution is really necessary we ran some numerical simulations. 100000 samples of the statistic defined in equation 1 were generated and used to construct the histogram shown in blue in figure 1. To calculate the sample means and variances for each sample of T we generated five standard normal random variables. 100 equally spaced bins between -7 and $+7$ were used when constructing the estimate of the probability density shown in figure 1. Any samples that fell outside this rage were added to the totals for the first and last histogram bins.

The red line in figure 1 shows the probability density function for the standard normal random variable. Meanwhile, the black line is the probability density for a Student t distribution with 4 degrees of freedom. The sampled distribution closely resembles the probability density function for the t-distribution. Like the t-distribution distribution the peak in the histogram is lower than that of the standard normal and the tails are fatter. If a log scale is used for y, as has been done in the right panel of figure 1, these differences in the tails are even more clear.

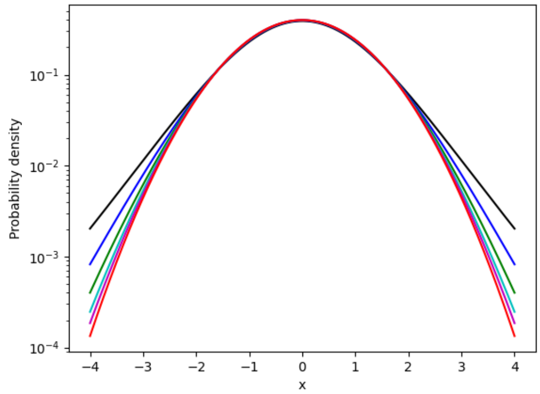

Fig. 2. Probability density functions for Student t-distributions with 10 (black), 20 (blue), 40 (green), 80 (cyan) and 160 (magenta) degrees of freedom. As the number of degrees of freedom increases the probability density function for the t-distribution comes to more closely resemble that of the standard normal distribution that is shown in red.

In figure 1 the probability density function for the t-distribution clearly falls within the confidence limit for our estimate of the probability density function for the statistic defined in equation 1. There is thus insufficient evidence to reject a null hypothesis that this statistic is a sample from this distribution. However, figure 1 also shows that probability density function for the normal distribution lies well outside of the confidence limits from our histogram. It is thus evident that our samples of T are not samples from a standard normal distribution. However, figure 2 shows, as you increase n Student's t-distribution gets progressively more and more similar to a standard normal distribution.

We can make this comparison between probability density functions more precise by introducing the following as a measure of the distance between two probability density functions, $p(x)$ and $q(x)$:

$$d = \int_{-\infty}^{\infty} |p(x) - q(x)| \mathrm{d}x \tag{2}$$

The black line in figure 3 was obtained by computing this integral numerically with $p(x)$ being the probability density function for the standard normal distribution and $q(x)$ being the probability density function for a t-distribution with number of degrees of freedom indicated on the y-axis. We used a grid of 1000 evenly spaced points between -10 and 10 when approximating the integral. You can see that the integral decreases monotonically as the number of degrees of freedom increases, which confirms that the distributions are becoming more similar.

To generate the black dashed line in the figure we computed the integral defined in 2 for the upper and lower bounds of the 90% confidence limit for the histogram that is shown in figure 1. The black line thus transforms the 90% confidence limit on our estimate of the distribution to a distance measured using equation 2. The figure shows that when the number of degrees of freedom is small the difference between the normal and t-distributions is larger

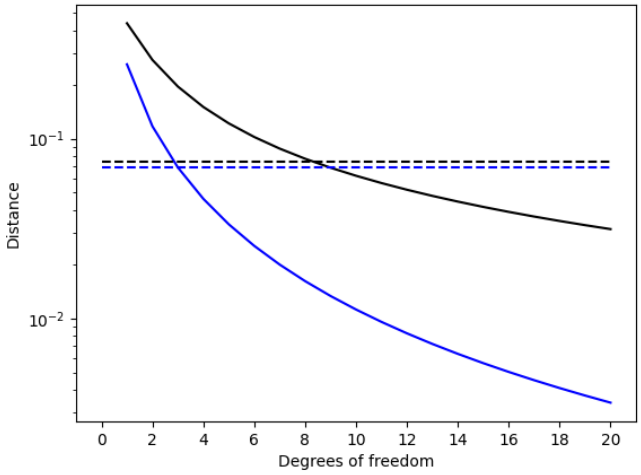

Fig. 3. The differences between the standard normal distribution and the Student t-distribution as a function of the number of degrees of freedom in the t-distribution. To generate the black line we computed these differences using equation 2, while the blue line shows the Kullback-Leibler divergence between the distributions. The dashed lines shows the values for these quantities that are obtained when they are computed for the upper and lower bounds on the confidence limit for the histogram from figure 1.

than the difference between the upper and lower bound on the estimate of the distribution in figure 1. However, once the t-distribution has 10 degrees of freedom or more the difference between the normal and t-distribution becomes smaller than this distance.

Equation 2 is not the only method we have for calculating the difference between two probability distribution. A technique that is commonly used in information theory involves the calculation of the Kullback-Leibler divergence, which is defined as follows:

$$D_{KL}(P||Q) = \int_{-\infty}^{\infty} p(x) \log\left(\frac{p(x)}{q(x)}\right) \mathrm{d}x$$

To generate the blue line in figure we computed D_{KL} for the standard normal and Student's t distributions with various numbers of degrees of freedom using the same numerical integration protocol that we used when evaluating equation 2. The blue dashed line then shows the D_{KL} value that we obtain from the upper and lower bounds of the confidence limit on the histogram that is shown in figure 1. You can see that when D_{KL} is used to measure the difference between the distributions the difference between the standard normal and the t-distribution is less than the difference between the upper and lower bounds on the estimate of the histogram in figure 1 once the t-distribution has 4 or more degrees of freedom.

3 Conclusion

The analysis in the previous sections have demonstrated that the test statistic defined in equation 1 is not a sample from a standard normal random variable when $n = 5$. However, our sampling has also not provided evidence that this random variable is not a sample from a Student t-distribution with four degrees of freedom.*

We then used various means for demonstrating that as the number of degrees of freedom increases Student's t-distribution comes to resemble a standard normal distribution ever more closely. The results we have obtained from these studies are contradictory when it comes to suggesting the value of n where a standard normal distribution can be safely used instead of the Student's t-distribution that the statistic defined by equation 1 is sampled from when performing a hypothesis test. We thus propose a simple solution: **always use the Student t-distribution** as this, after all, is the distribution that equation 1 is sampling. Modern computational statistics packages make it straightforward to calculate the probability density function for the Student t-distribution with any number of degrees of freedom. The only reason students were ever told to use the standard normal approximation when n is large is that it is not possible to print out the t-distributions in the statistical tables for all possible values for the number of degrees of freedom.

*This second result is not surprising, the textbook theory tells us that the test statistic in equation 1 **is** a sample from this distribution. However, it is important to remember that we can **never** prove this fact by sampling.